

E-COMMERCE ORDER MANAGEMENT SYSTEM

1. Aim

To create **Seller and Inventory tables** and manage seller details, product information, stock availability and inventory analysis using SQL.

2. Seller Table

The Seller table stores information about product suppliers.

Column	Data Type	Constraint
Seller_ID	INT	Primary Key
Seller_Name	VARCHAR(100)	NOT NULL
Seller_Email	VARCHAR(100)	UNIQUE, NOT NULL
Seller_Phone	VARCHAR(15)	NOT NULL

SQL Code

```
CREATE TABLE Seller (  
    Seller_ID INT PRIMARY KEY,  
    Seller_Name VARCHAR(100) NOT NULL,  
    Seller_Email VARCHAR(100) UNIQUE NOT NULL,  
    Seller_Phone VARCHAR(15) NOT NULL  
);
```

3. Inventory Table

The Inventory table stores product stock information and tracks available stock.

Column	Data Type	Constraint
Inventory_ID	INT	Primary Key
Product_ID	INT	UNIQUE, NOT NULL

Stock_Quantity INT NOT NULL, CHECK

SQL Code

```
CREATE TABLE Inventory (  
    Inventory_ID INT PRIMARY KEY,  
    Product_ID INT UNIQUE NOT NULL,  
    Stock_Quantity INT NOT NULL CHECK (Stock_Quantity >= 0)  
);
```

4. Sample Records

Seller Records

```
INSERT INTO Seller  
VALUES  
(1, 'ABC Traders', 'abc@gmail.com', '9876543210'),  
(2, 'Tech World', 'tech@gmail.com', '9876543211'),  
(3, 'Smart Store', 'smart@gmail.com', '9876543212'),  
(4, 'Global Mart', 'global@gmail.com', '9876543213');
```

Inventory Records

```
INSERT INTO Inventory  
VALUES  
(101, 1, 25),  
(102, 2, 5),  
(103, 3, 0),  
(104, 4, 15);
```

5. CRUD Operations

CREATE – Add New Seller

```
INSERT INTO Seller  
VALUES (5, 'New Store', 'newstore@gmail.com', '9876543214');
```

READ – Display Sellers

```
SELECT * FROM Seller;
```

UPDATE – Update Seller Contact

```
UPDATE Seller
```

```
SET Seller_Phone = '9999999999'  
WHERE Seller_ID = 1;
```

DELETE – Delete Seller

```
DELETE FROM Seller  
WHERE Seller_ID = 5;
```

6. Track Product Availability

Available Products

```
SELECT *  
FROM Inventory  
WHERE Stock_Quantity > 0;
```

Out of Stock Products

```
SELECT *  
FROM Inventory  
WHERE Stock_Quantity = 0;
```

Low Stock Products

```
SELECT Product_ID, Stock_Quantity  
FROM Inventory  
WHERE Stock_Quantity < 10;
```

Update Inventory After New Stock Arrival

```
UPDATE Inventory  
SET Stock_Quantity = Stock_Quantity + 20  
WHERE Product_ID = 2;
```

Remove Discontinued Inventory

```
DELETE FROM Inventory  
WHERE Product_ID = 3;
```

7. Inventory Analysis

Total Products Available

```
SELECT COUNT(*) AS Total_Products_Available  
FROM Inventory  
WHERE Stock_Quantity > 0;
```

Products Out of Stock

```
SELECT COUNT(*) AS Out_Of_Stock
FROM Inventory
WHERE Stock_Quantity = 0;
```

Highest Stocked Product

```
SELECT Product_ID, Stock_Quantity
FROM Inventory
WHERE Stock_Quantity = (
    SELECT MAX(Stock_Quantity)
    FROM Inventory
);
```

Average Inventory Quantity

```
SELECT AVG(Stock_Quantity) AS Average_Inventory
FROM Inventory;
```

8. ER Relationship

Seller → Product

One seller can supply multiple products. The document also states that a product can be supplied by one or multiple sellers.

Product → Inventory

Each product has inventory information and inventory tracks available stock.

```
SELLER
|
| supplies
|
PRODUCT
|
| has
|
INVENTORY
```

9. Final Submission

1. Seller Table Design
2. Inventory Table Design

3. ER Relationship Diagram
4. SQL Table Creation Script
5. Sample Seller & Inventory Records
6. CRUD Operations
7. Inventory Analysis Queries